

Scatterplots and Linear Regression

For this assignment, you will be using the data set **car_data.csv**. This data set is a modified version of the data set imports-85.data from <https://archive.ics.uci.edu/ml/datasets/automobile>. There are two columns in this data set. One is the weight of the vehicle in pounds, and the other is the highway miles per gallon.

1. Make a scatterplot of the highway miles per gallon (y -axis) versus the weight (x -axis).
2. Based on your plot, what is the general trend of how highway miles per gallon varies with the weight?
3. If you were to build a linear model using this data to predict highway miles per gallon from weight, would you expect the slope to be positive or negative? Explain.
4. If the slope of a linear model predicting highway miles per gallon from the weight, interpret the meaning of the slope being -0.05 .
5. Write code to add a line to the graph you made in problem (1). Adjust the slope and y -intercept of this line until you think you have found the line that best fits the data. Record the slope and y -intercept.
6. Use Python to find the best-fit (least-squares) line. The Scikit-learn package is a good choice to use for this.
7. Find the root mean squared error (RMSE) of the prediction line you found in problem (5) and the actual best-fit line found in problem (6). How do these compare? Is this what you would expect?
8. Use the best-fit line in problem (6) to predict the highway mpg of a car that weighs 3200 pounds.